Stick-together calculations

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find a common final velocity using total momentum.

SUCCESS CRITERIA - I can...

1. Add the initial momenta with signs.
2. Use the combined mass after sticking.
3. Check the result against the direction of total momentum.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Fragments of masses **3** and **5 kg** separate from rest. The **3 kg** piece moves at **+10 m/s**. **Find the other velocity** and total kinetic energy.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $v_f = (m_1 v_{1,i} + m_2 v_{2,i})/(m_1 + m_2)$.
Sticking collision in one dimension with negligible external impulse. v_f is the shared final velocity.

Worked reasoning: A **2 kg** cart at **+3 m/s** sticks to a **1 kg** cart at rest. Total momentum **6 kg·m/s** divided by total mass **3 kg** gives final velocity **+2 m/s**.

Your turn: Mark the conserved quantity on a before-and-after sketch. **Explain why masses** are added in the final denominator.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two carts stick. **Identify which masses belong** in the final momentum expression and why.

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Apply the physics

A **2 kg** cart at **+5 m/s** sticks to a **3 kg** cart at **-1 m/s**. **Find final velocity** and kinetic energy transformed. Organize the known quantities on the before-and-after diagram.

Before	After

Label objects and signed velocities.

Marzano check - circle: ① ② ③

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Continue down each column.

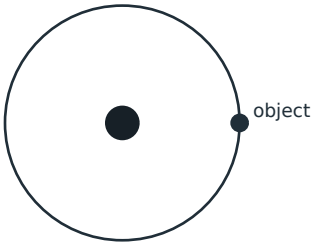
05 Explain your reasoning

A student averages the two starting velocities without considering mass. **When would** that happen to work?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

At fixed mass and speed, circle radius becomes **4** times as large. **Compare acceleration** and net force using equations.
Label velocity and inward acceleration on the circular-path diagram.



Marzano check - circle: 1 2 3

07 Exit check

A **0.20 kg** projectile at **30 m/s** sticks to a **1.80 kg** cart at rest. **Find common speed** and fraction of initial kinetic energy remaining. Organize the known quantities on the before-and-after diagram.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **1 kg** cart at **+6 m/s** sticks to a **2 kg** cart at **-3 m/s**. **Find final velocity**. If the second velocity were **-1 m/s** instead, find the new result. Organize the known quantities on the before-and-after diagram.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3